Practice workbook · Identify brain regions and match them to their functions

Labelling or locating the cerebral hemispheres, cerebellum, medulla, hypothalamus and pituitary, and pairing each with functions such as balance, memory, breathing or temperature control.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATCH-BRAIN-AREA-TO-FUNCTION

1. Question 1 · part b · 4 marks

June 2021 Paper 2

Worked-example retry · Attempt part b only.

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1 The diagram shows a section through a human brain.



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(a) Using lines, label these structures on the diagram.

- cerebral hemisphere
- cerebellum
- pituitary gland
- hypothalamus

(4)

(b) State which structure in part (a) is responsible for each of these functions.

(4)

Function	Structure
control of muscle tone and posture	
intelligence and thought	
monitoring of body temperature	
release of ADH	

(Total for Question 1 = 8 marks)

1. Question 1 · part b · 4 marks · Working and answer

Working and answer

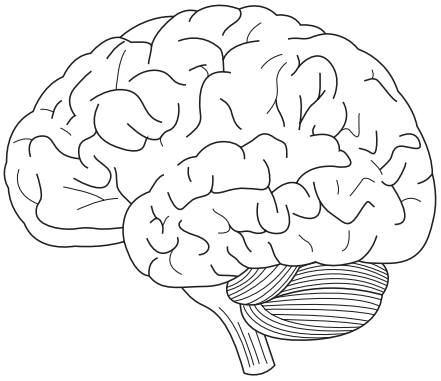
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2. Question 7 · part a · 3 marks

June 2019 Paper 1

Further practice · Attempt part a only.

7 (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

(3)

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

(1)

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(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

(1)

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2. Question 7 · part a · 3 marks · Working and answer

Working and answer

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3. Question 6 · part b · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

The B-amyloid is formed when an essential protein called amyloid precursor protein (APP) is not broken down correctly.

(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

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2 .....

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3. Question 6 · part b · 1 mark · Working and answer

Working and answer

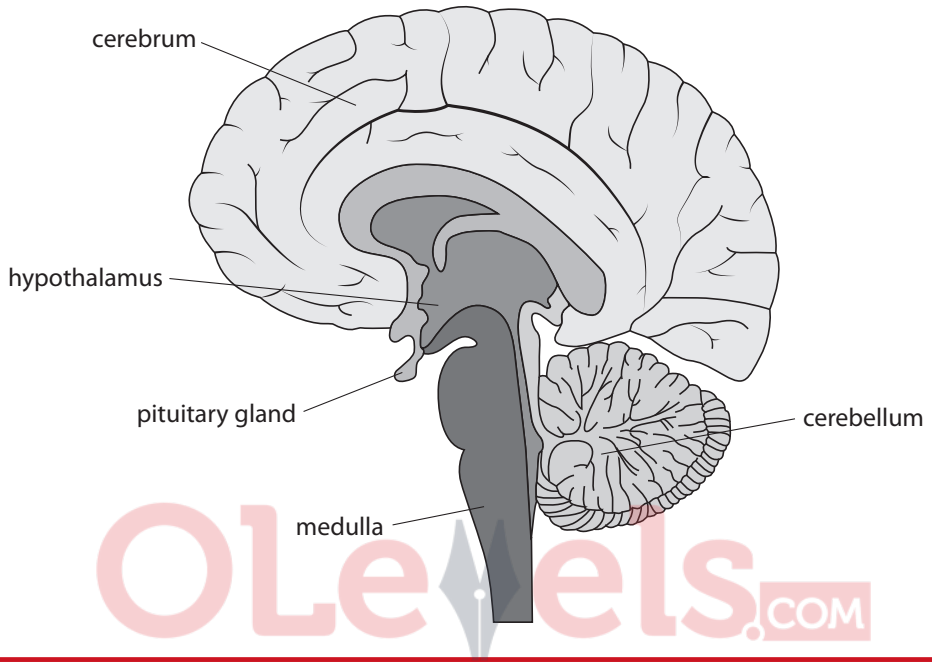
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4. Question 7 · part a.i · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

7 The diagram shows parts of the human brain.



(a) (i) The table lists parts of the brain and some functions of the brain.

Complete the table by putting a tick (✓) to show if the function is correct for each part.

(3)

Part	Function	Tick
medulla	controls heart rate and breathing	
pituitary gland	involved in language	
hypothalamus	responsible for osmoregulation	
cerebellum	controls voluntary movements	

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4. Question 7 · part a.i · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

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(ii) The cerebrum is divided into the left and right cerebral hemispheres.

Each hemisphere contains different areas. One of these is the motor area.

State the function of the motor area.

(1)

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(b) Nerve cells deep inside the brain produce the neurotransmitter dopamine.

Parkinson's disease is a brain disorder that destroys these nerve cells.

Explain why the destruction of these nerve cells causes the symptoms associated with Parkinson's disease.

(3)

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(Total for Question 7 = 7 marks)

4. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

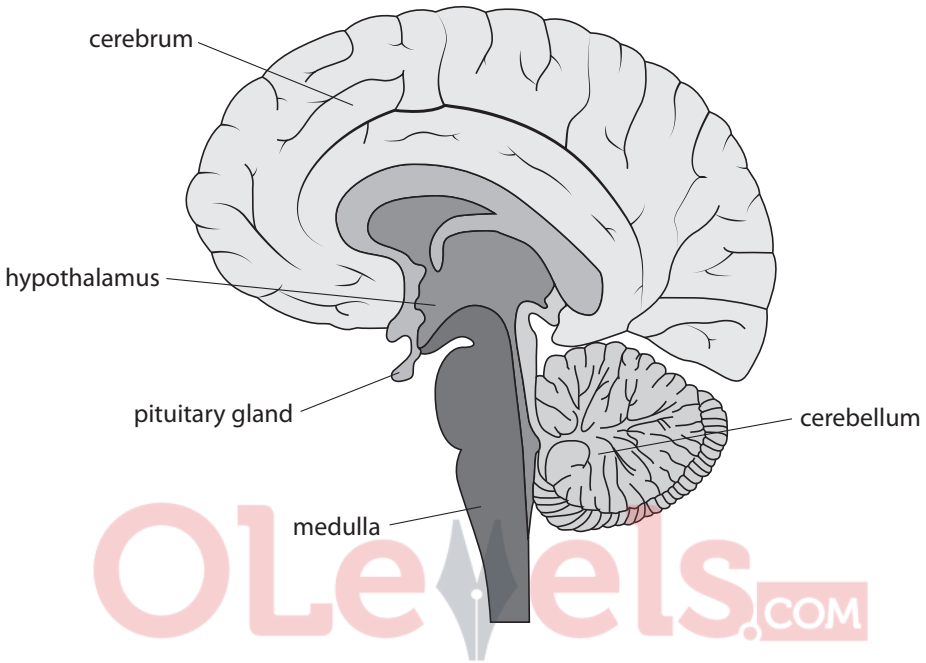
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5. Question 7 · part a.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.ii only.

7 The diagram shows parts of the human brain.



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Complete the table by putting a tick (✓) to show if the function is correct for each part.

(3)

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medulla	controls heart rate and breathing	
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5. Question 7 · part a.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.ii only.

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(ii) The cerebrum is divided into the left and right cerebral hemispheres.

Each hemisphere contains different areas. One of these is the motor area.

State the function of the motor area.

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(3)

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(Total for Question 7 = 7 marks)

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Working and answer

6. Question 6 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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6. Question 6 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

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(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

6. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

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7. Question 1 · part a · 4 marks

June 2021 Paper 2

Further practice · Attempt part a only.

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1 The diagram shows a section through a human brain.



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(a) Using lines, label these structures on the diagram.

- cerebral hemisphere
- cerebellum
- pituitary gland
- hypothalamus

(4)

(b) State which structure in part (a) is responsible for each of these functions.

(4)

Function	Structure
control of muscle tone and posture	
intelligence and thought	
monitoring of body temperature	
release of ADH	

(Total for Question 1 = 8 marks)

7. Question 1 · part a · 4 marks · Working and answer

Working and answer

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8. Question 3 · part a · 1 mark

November 2021 Paper 2

Further practice · Attempt part a only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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8. Question 3 · part a · 1 mark · Working and answer

Working and answer

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9. Question 3 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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9. Question 3 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

9. Question 3 · part d · 1 mark · Working and answer

Working and answer

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